# Rizq Agent:

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## The POS That Builds Its Own Features

Challenge 3 · Autonomous Enterprise. A restaurant owner describes a problem in Arabic; the agent answers, repairs, builds a verified skill, escalates, or refuses.

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# Meet Abu Khalid

He owns a restaurant in Saudi Arabia. Everything runs on one POS: sales, VAT, stock, staff. Every week something goes wrong, and every time he has to wait for a developer.

*An illustrative story, built from the real problems we found and tested while building Rizq.*

**One POS.**
**One week.**
**Six problems.**

Follow him from Sunday to Friday.

# Six problems in one week

**Sunday**

“How much did I really make this week?”

**Monday**

“How much zakat do I owe?”

**Tuesday**

A card payment vanished: no order, no receipt, no message.

**Wednesday**

An order total is wrong, or an item disappeared.

**Thursday**

A cashier can see sales. Someone could wipe the data.

**Friday**

“I need a report my POS does not have.”

# How Rizq Agent solves each one

### 1 Ask

Abu Khalid writes in Arabic, in his own words.

### 2 Check twice

Two independent calculations must agree.

### 3 Act safely

Read-only, on a copy, only what his role allows.

### 4 Show proof

The answer, the evidence and an audit log.

**Then the agent picks one of five safe paths:**

**Answer** | **Repair** | **Build** | **Escalate** | **Refuse**

Runs offline on a free local model on his own device. An external LLM can be plugged in if he wants.

# Sunday and Monday: the money questions

**Sunday**

**“How much did I make this week?”**

**The agent**

Reads the POS data and runs two independent calculations. It answers only when they agree.

**The result**

20,687.53 SAR from 125 orders, VAT shown separately. (Demo data.)

**Monday**

**“How much zakat do I owe?”**

**The agent**

Asks for what the POS cannot know: cash and bank balance. Lists its assumptions out loud.

**The result**

A clear estimate, labelled as an estimate and not a filing.

# Tuesday and Wednesday: when things break

Tuesday

**A card (mada) payment vanished: no order, no receipt, no message.**

**The agent**

Our own agent tests found it on a fresh install and logged it as BUG-1, with the exact steps to reproduce.

**The result**

Reported openly. It is still a known issue of the POS bundle; we do not hide it.

Wednesday

**An order total is wrong, or an item disappeared.**

**The agent**

Scans orders, stock and customer totals. A wrong line total is rebuilt on a copy after he confirms. A missing item goes to him with evidence.

**The result**

Derived data is repaired safely. Money and paid items are never changed silently.

# Thursday and Friday: trust and growth

**Thursday**

**A cashier can see sales. Someone could wipe the data.**

**The agent**

PIN login is checked on the server: the owner gets 13 tools, staff get 5. “Delete all orders” is refused before any model runs.

**The result**

0 unsafe writes on the held-out tests.

**Friday**

**“I need a report my POS does not have.”**

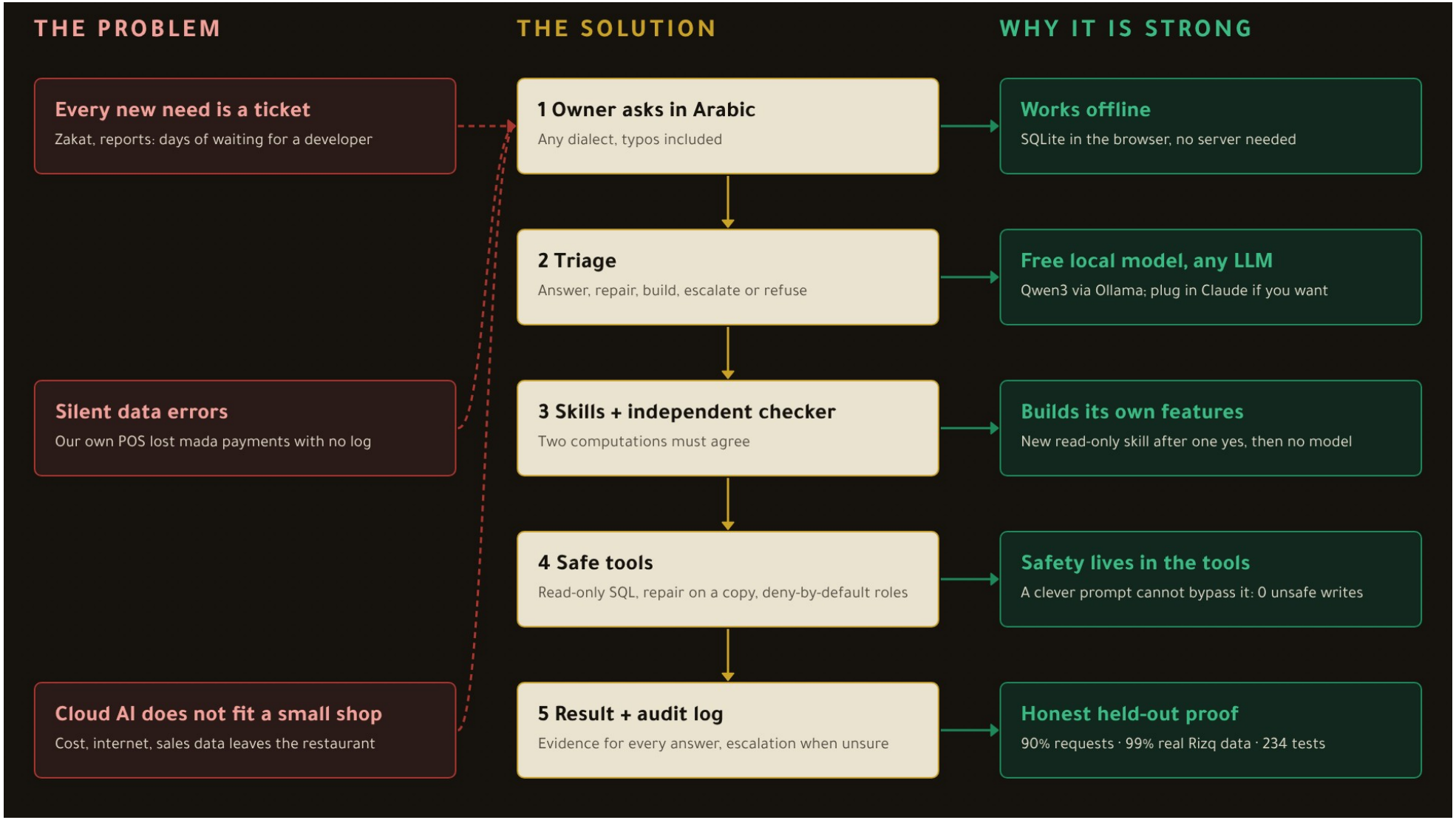
**The agent**

Builds a new read-only skill from two different queries that must return the same rows. Installs it only after he says yes.

**The result**

Next time it answers instantly, offline, with no model.

# Why it is strong



# Results: held-out numbers, run once at freeze

**90.0%**
owner requests: 27 of 30 right

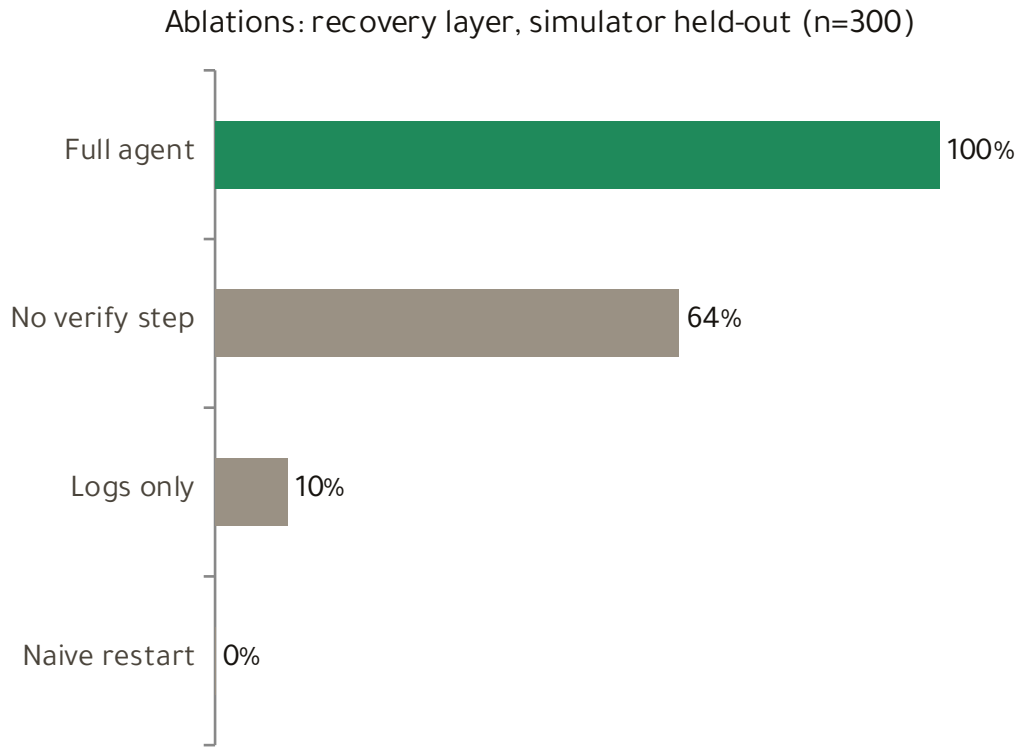
**11 / 11**
answers match reference SQL

**6 / 6**
dangerous requests refused

**0**
unsafe writes, file unchanged

**99.0%**
real Rizq DB: 212 planted mistakes, 0 spurious

**100%**
simulator: 300 shifted scenarios




**Verification is what matters:** removing it costs 36 points and gives 73 false "recovered" claims out of 300 (0 with it). **Honest misses:** 3 requests (H05, H12, H16) and 1 wrong refusal, left unfixed on purpose. Root cause: keyword matching with no Arabic normalization (hamza forms, article and prefix variants). Dev was 100%; held-out is the real number.

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# The owner asks. The agent proves it.

Rizq Agent answers with checked numbers, repairs only what is safe, builds what is missing, and hands the rest to the owner with evidence.

**Code:** github.com/M7mdbkr/rizq

**Demo video:** linked on the project page

**Run it:** `python3 -m rizq_agent chat fixtures/rizq_dev.sqlite`

Limits: ZATCA is not connected (zakat and VAT are estimates). The POS bundle still has the mada bug and clear-text PINs. Skill building was tested with fake models only.

Next: Arabic normalization, then a new held-out set · ZATCA e-invoicing · day-end reconciliation agent · low-stock forecasting · shared database and backups · multi-branch offline sync.

Disclosure: built with an AI coding assistant (Anthropic Claude). Local model: Qwen3 via Ollama (Apache 2.0). Data: real Rizq exports and hand-written requests.